

Japanese Perceptual Epenthesis is Modulated by Transitional Probability

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Abstract

Perceptual epenthesis is the perception of illusory vowels in consonantal sequences that violate native phonotactics. The consensus has been that each language has a single, predictable candidate for perceptual epenthesis, that vowel which is most minimal (i.e., shortest and/or quietest). However, recent studies have shown that alternate epenthetic vowels can be perceived when the perceptual epenthesis of the minimal vowel would violate native co-occurrence restrictions. We propose a potential explanation for these observed patterns: speech perception, and thus also vowel perceptual epenthesis, is modulated by transitional probability whereby epenthetic vowels must conform to the language specific expectations of the listener. To test this explanation, we present two experiments examining perceptual epenthesis of two Japanese vowels—/u/ and /i/—against their transitional probability in CV sequences. In Experiment 1, Japanese listeners assigned VCCV tokens to VCuCV and VCiCV categories. In Experiment 2, participants discriminated VCCV tokens from VCuCV and VCiCV tokens. The results show that sequences where /i/ is transitionally probable are more likely to elicit /i/ perceptual epenthesis.

Keywords

Perceptual epenthesis, transitional probability, surprisal, entropy, Japanese

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Introduction

A characteristic feature of Japanese phonology is its restrictive phonotactic system, which disallows non-homorganic consonant clusters. When Japanese listeners are exposed to non-homorganic consonant clusters, they sometimes perceive an illusory /u/, which serves to perceptually repair the sequence so that it conforms to Japanese phonotactics (e.g., /ebzo/ → /ebuzo/; Dupoux et al., 1999). This process is known as perceptual epenthesis. It has been argued that the vowel /u/ is the candidate for perceptual epenthesis in this context—as opposed to other vowels—because Japanese /u/ has the shortest duration of Japanese vowels (e.g., Arai et al., 2001; Campbell, 1992, 1999; Sagisaka & Tohkura, 1984) and has devoiced allophones (e.g., Fujimoto 2015), making it the most “phonetically minimal element of the language” (Dupoux et al., 2011, p. 200). If the epenthesis candidate must be the most phonetically minimal vowel, then each language should only show a single candidate for perceptual epenthesis (unless two or more vowels are equally minimal).

However, recent studies have shown that languages can have multiple epenthetic vowels (Durvasula & Kahng, 2015; Durvasula et al., 2018). One such study found that Korean listeners are significantly more likely to perceive an alternate epenthetic vowel (/i/) in sequences where the production of the dominant vowel (/ɪ/) would violate Korean phonotactics (Durvasula & Kahng, 2015). We interpret these findings to suggest that phonotactic violations are frequently misperceived because they are *unexpected* by the listener. In the following, we explore the role of listener expectations through the measures of Surprisal and Entropy—two analytical tools in Shannon’s (1948) Information Theory—which serve as proxies for listener expectations (see the discussion in Section 1.2). We propose that speakers often misperceive non-native sequences of speech that are unexpected as sequences that are both perceptually similar and expected in their native language (L1) (Kilpatrick et al., 2019), and we test this proposal in two perceptual experiments (Experiment 1: Categorization reported in Section 2; Experiment 2: Discrimination reported in Section 3), which investigate the influence of transitional probability on perceptual epenthesis. Experiment 1 is a cross-language mapping task whereby Japanese listeners categorize both licit ([eCupo] and [eCipo]) and illicit ([eCpo]) tokens to legal Japanese categories (/eCupo/ or /eCipo/) and rate how well they “fit” to the assigned category. Experiment 2 is an AXB discrimination task. The results show that transitional probability provides a better explanation for the distribution of perceived epenthetic vowels than phonotactic constraints alone.

1.1 Epenthetic vowels in Japanese

If phonetic minimality is the single factor that determines the vowel quality in perceptual epenthesis, then we would expect that listeners will only experience exactly one epenthetic candidate for their respective languages. However, the Korean case examined by Durvasula and Kahng (2015) shows that phonetic minimality is not the only determining factor in perceptual epenthesis. We propose that their findings are due to the extremely low transitional probability of certain /Ci/ sequences due to Korean co-occurrence restrictions: unattested sequences are mapped to sequences that are both perceptually similar and transitionally probable. We might therefore predict more than one epenthetic vowel to be available in a given language, depending on this factor; each language can in principle have any number of epenthetic vowels, provided that such candidates are both suitably minimal and transitionally probable.

Existing research has explored the possibility of multiple epenthetic vowels in Japanese. Japanese has co-occurrence restrictions against */tu/ and */du/. In loanwords, /t/ and /d/ are typically assimilated to /to/ and /do/ when they occur word-finally or in the coda position of non-homorganic

consonant clusters (Ito, 1989). Studies have shown that Japanese listeners experience neither /o/ nor /u/ perceptual epenthesis in consonant clusters following alveolar stops (Ito, 1989; Mattingley, Hume & Hall, 2015; Monahan et al., 2009). Unlike the Korean case noted above, the co-occurrence restriction limiting the preferred candidate does not elicit /o/ perceptual epenthesis, suggesting that /o/ is unsuitable for perceptual epenthesis in Japanese. Another study found that Japanese listeners overwhelmingly identified an epenthetic /i/ in consonant clusters following /dz/,¹ but were more likely to select /u/ in all other examined contexts (Mattingley et al., 2015), while yet other studies have shown that the vowels flanking the consonant cluster can influence the quality of the epenthetic vowel (Dupoux et al., 2011; Guevara-Rukoz et al., 2017). These studies have found a “vowel copy” effect whereby the acoustic cues of the flanking vowels carry over the consonant. The vowel copy effect was found to be stronger when the medial consonant is [h] compared to when it is a plosive or when the flanking vowels are identical (Guevara-Rukoz et al., 2017; cf. Kawahara, 2004, for a similar loanword adaptation pattern).

Recent studies have outlined a possible relationship between Japanese perceptual epenthesis and high vowel devoicing (e.g., Cutler, Otake & McQueen, 2009; Kilpatrick et al., 2018). Japanese listeners are more likely to treat non-homorganic consonant clusters as potential realizations of parts of existing Japanese words if the consonants are voiceless (Cutler, Otake & McQueen, 2009). Japanese high vowels systematically undergo devoicing when they occur either between voiceless consonants or following a voiceless consonant at the end of a word (Fujimoto, 2015; Kubozono, 2015).

$$[V_{[+high]}] \rightarrow \left[\begin{smallmatrix} V \\ \circ \end{smallmatrix} \right] / \left[\begin{smallmatrix} C \\ \circ \end{smallmatrix} \right] - \left[\begin{smallmatrix} C \\ \circ \end{smallmatrix} \right]$$

$$[V_{[+high]}] \rightarrow \left[\begin{smallmatrix} V \\ \circ \end{smallmatrix} \right] / \left[\begin{smallmatrix} C \\ \circ \end{smallmatrix} \right] - \#$$

It has been argued that devoiced vowels that follow stops are different, in some sense, to devoiced vowels that follow fricatives and affricates, and some contend that the latter are in fact deleted (see Shaw & Kawahara, 2018a).² Surface-level deletion in these contexts would mean that Japanese tolerates non-homorganic consonant clusters and Japanese listeners would be unlikely to experience perceptual epenthesis when exposed to such sequences. However, contrary to this expectation, studies have shown that Japanese listeners experience more salient illusory vowels when the epenthetic context is preceded by a voiceless fricative (Kilpatrick et al., 2018). The researchers propose that this is because perceptual illusions are more likely to occur when the input and the illusion are more similar. The turbulent aperiodic energy of the voiceless fricative masks the acoustic cues of the devoiced vowel so the perceptual distance between the violation and the repair target is smaller.

Japanese vowel devoicing is particularly relevant to the relationship between predictability and perceptual epenthesis because if the identity of the vowel is predictable from its context, listeners can recover the intended message, even when the speaker invests little or no resources in the production of devoiced high vowels. Japanese vowel devoicing occurs habitually in several high-frequency words and suffixes (the affirmative present tense suffix /-masu/; the past tense suffix /-šita/; the imperative suffix /-šite/; and copula /desu/ (Kilpatrick et al., 2016; Maekawa, 1989; Maekawa & Kikuchi, 2005; Tsuchida, 1997). These results are consistent with the conclusion that some sort of frequency measure shapes the patterns of perceptual epenthesis. Studies have shown that Shannon’s (1948) Entropy calculations can be used to predict the devoicing behavior of high

Table 1. Tokens used in Experiments 1 and 2.

Licit /u/	Licit /i/	Illicit
/egupo/	/egipo/	/egpo/
/eʃupo/	/eʃipo/	/eʃpo/
/etʃupo/	/etʃipo/	/etʃpo/

vowels whereby CV contexts with a low Entropy (Section 1.2) are more likely to elicit vowel devoicing (or deletion, as proposed by Whang, 2018). Building on previous research that explores the general influences of predictability on sound patterns (e.g., Aylett & Turk, 2004, 2006; Bell et al., 2009; Cohen Priva, 2012, 2015; Hume, 2016), we propose that this behavior is the result of a trade-off between resource cost and communicative accuracy whereby predictable sequences of sounds are produced less robustly by speakers (Zipf, 1948). We discuss the theory behind this proposal in the following section.

1.2 Transitional probability

In linguistic research, transitional probability has mainly been used to predict and account for the behavior of speakers in terms of the degree of effort invested in the production of words and sequences of speech of varying predictability. Research on several languages has shown that segments, syllables, and words that are more predictable tend to be produced less robustly (e.g., Aylett & Turk, 2004, 2006; Bell et al., 2009; Cohen Priva, 2012, 2015; Hume, 2016; Zipf, 1948). Shaw and Kawahara (2017) examined this phenomenon at the level of phonotactics—rather than at the word level—to explore the effect of transitional probability on vowel duration in Japanese. They calculated Surprisal and Entropy at the level of the CV mora to show that transitional probability influences the duration of vowels. Generally, the more likely it is that a Japanese consonant and vowel co-occur in CV sequences (i.e., the lower the Surprisal), the shorter the vowel. Several recent studies have made use of Shannon’s (1948) Entropy and Surprisal (e.g., Cohen Priva, 2012; Hume, 2016) to examine the effects of predictability on robustness of speech production. Surprisal is calculated as the negative log (base 2) of the transitional probability between a consonant and a vowel.

$$\text{Surprisal } (v | C) = -\log_2(v | C).$$

This calculation provides a measure of how “surprising” or (un)predictable a given vowel is given the preceding consonant. Consonant-conditioned vowel Entropy, represented as H , is the weighted average of the Surprisal of each vowel given the preceding consonant. Vowel Entropy provides a measure of uncertainty for the preceding consonants; it is weighted averages of Surprisal values, reflecting the number of distinct competitors based on type frequencies across the lexicon (Shaw & Kawahara, 2017).

$$\text{Entropy: } H(C) = -\sum(x | C) * \log_2 p(x | C), \text{ where } x \in V$$

The present study calculates Surprisal and Entropy using counts from the *Corpus of Spontaneous Japanese*, which is one of the largest corpora in Japanese (Maekawa, 2003). The frequency counts were obtained using its core portion (CSJ-RDB), which contains more than 300,000 vowel tokens

Table 2. Relevant co-occurrence counts from the CSJ-RDB.

	/g/	/ʃ/	/tʃ/
/a/	19,439	1,740	478
/i/	1,168	26,530	7,219
/u/	2,087	1,033	23
/e/	2,258	14	15
/o/	5,441	1,715	995
Total	30,393	31,032	8,730

Table 3. Surprisal calculation measured in bits of information; the less likely a combination, the higher the Surprisal.

	/g/	/ʃ/	/tʃ/
/a/	0.64	4.16	4.19
/i/	4.70	0.23	0.27
/u/	3.86	4.91	8.57
/e/	3.75	11.11	9.18
/o/	2.48	4.18	3.13

produced by more than 200 speakers. The following experiments examine L1 Japanese listeners' perception of non-homorganic consonant clusters. Clusters feature [g], [ʃ], and [tʃ] in the preceding position and [p] in the following position (the full list of tokens is presented in Table 1).

Of the three consonants examined, the CSJ-RDB reports that only /g/ occurs more frequently with /u/ than with /i/ in CV sequences. The CSJ-RDB uses a separate phonetic symbol for /g/, /ʃ/, and /tʃ/ when they occur before /i/ compared to when they occur before other vowels to express coarticulatory palatalization due to /i/. In the following, these have been combined, which allows us to calculate how often each vowel is combined with each consonant (Table 2) and, consequently, the Surprisal of each vowel after each consonant (Table 3).

Based on Surprisal, Shannon's Entropy can be calculated (Table 4). Entropy provides a measure of the amount of information a context carries. This paper deploys Surprisal and Entropy calculations based on the two high vowels /i/ and /u/ for the following reasons (for a recent discussion of different choice spaces for calculating predictability, see Shaw & Kawahara, 2018b). Whang (2018) employed an approach that considers only high vowel transitional probability and showed that such an approach can be used to account for vowel devoicing behavior. This is particularly relevant given that research has shown the importance of devoicing contexts on perceptual epenthesis (Kilpatrick et al., 2018). In addition, Monahan et al.'s (2009) study has shown that [o] does not pattern like [u] in terms of perceptual epenthesis. There is thus a good body of evidence suggesting that high vowels are privileged as candidates for perceptual epenthesis.

The above calculations are based upon CV co-occurrence frequencies, which is likely the most relevant unit of speech given the saliency of the mora as a prosodic unit. Indeed, Shaw and Kawahara (2017) have shown that Entropy calculations based on CV transitional probability can affect vowel duration, but it is unlikely that transitional probability influence is limited only to the consonant preceding the epenthetic context. However, Entropy cannot be calculated on sequences larger than CV mora in the current case because the CSJ-RDB reports zero instances of /etʃu/ and /tʃup/. In the following section, we report our first experiment, which tests Japanese listeners'

Table 4. High vowel Entropy calculations. The lower the Entropy, the more predictable the following vowel.

	/g/	/j/	/tʃ/
High vowel Entropy	0.45	0.36	0.25

Table 5. Response categories and their hiragana representation.

Category	/egipo/	/egupo/	/ejipo/	/ejupo/	/etʃipo/	/etʃupo/
Hiragana	えぎぼ	えぐぼ	えしぼ	えしゅぼ	えちぼ	えちゅぼ

perception of the tokens listed in Table 1 and asks that they categorize and report how well tokens fit to an /eCipo/ or an /eCupo/ category.

2 Experiment 1: Categorization

2.1 Method

2.1.1 Participants. Thirty-one students (female = 19; age 19–28, $M = 21$, $SD = 1.8$) from the Mita campus of Keio University (Tokyo) were recruited by word of mouth. Participants had limited exposure to languages other than Japanese; all were born in Japan to Japanese-speaking parents; and none had lived outside of Japan for longer than one month. All participants had undertaken six years of compulsory English education through middle and high school and many had undertaken additional English study. A small number of students indicated that native English speakers contributed to their English instruction; however, most reported that their English instruction was facilitated entirely by native Japanese speakers. Participants took approximately 60 minutes to complete the experiments and relevant paperwork. One participant failed to complete the experiments and was excluded from the analysis. Participants were compensated JP¥1500 for their time. Participants were informed that the present study had received ethics approval from the University of Melbourne and gave written consent prior to commencing the experiment.

2.1.2 Procedure. Experiment 1 consisted of a categorization experiment in which participants were exposed to the nine /eCVpo/ and /eCpo/ strings presented in Table 1. Participants were asked to categorize each string into one of six /eCVpo/ categories, which were presented to the participants as on-screen buttons with hiragana labels (as shown in Table 5). The experimental strings were drawn randomly from a library of 81 tokens; three repetitions of each of the nine strings from each of the three speakers. Each token was presented to participants three times for a total of 243 trials. Participants were given 3500 ms to respond to both the categorization and goodness-of-fit tasks. If they failed to respond within that time limit, the presentation would move to a new token and the missed trial would be replayed at a random time during the remainder of the experiment. Participants were not given any information regarding the language in which the strings had been produced but given that experiments were conducted by a non-Japanese researcher, participants potentially assumed that strings would be produced by non-Japanese speakers (perhaps even specifically by speakers of English), despite the on-screen hiragana response “buttons.” Categorization response times were measured from the offset of the presentation of each token (inclusive of the 10 ms ramp-out). Upon assigning each token a category, participants would be presented with

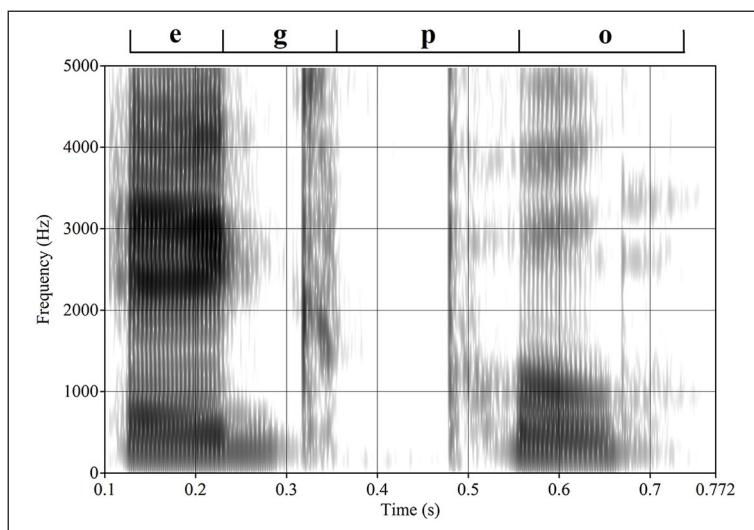


Figure 1. Spectrogram of [egpo] produced by one of the three Australian English speakers.

the exact same token and asked to assign it a goodness-of-fit rating to the assigned category along a Likert scale from 1 to 7. The 7 on the Likert scale was labeled with 同じ (identical) and the 1 was labeled with 違う (different) to indicate that a high score represented a good fit.

The experiment was conducted in a quiet room located at the Mita Campus of Keio University. Each participant first completed a detailed language background questionnaire and read an instructional handout, both of which were presented in Japanese. The experiment was presented individually to participants over noise canceling headphones from a MacBook Pro using PsyScope X, following the completion of a practice test. Participants used an optical mouse to enter their responses (see Table 5). The practice task typically lasted less than a minute before the participants indicated that they understood the test instructions.

2.1.3 Stimuli. The stimuli for Experiment 1 are shown in Table 1. The medial /CV/ and /C/ sequences were embedded in a nonce frame constructed from /e/, /p/, and /o/ because these phonemes occur infrequently in Japanese and thus limit unintended variability to the dataset beyond those factors deliberately manipulated. Tokens were constructed into either licit /eCVpo/ or illicit /eCpo/ sequences. Vowels presented in the epenthetic position in licit sequences were either /i/ or /u/ and consonants preceding the epenthetic position were /g/, /ʃ/, or /tʃ/. None of the targets are existing Japanese words. Three female Australian English (AusE) speakers produced three tokens of each target with initial stress, in citation (list initial and final tokens were excluded). Recordings were conducted in the Horwood recording studio at the University of Melbourne, using a Charter Oak E700 condenser microphone and Tieline Commander G3 codec protocol recording in mono with 16-bit resolution. Tokens were excised using a *Praat* script, which also enveloped tokens with a 10 ms ramp-in and a 10 ms ramp-out. All tokens were checked by the first author after excising. Particularly careful attention was given to [egpo] tokens because AusE disallows /gp/ clusters, although these sequences do occur across word boundaries, as in “big pot.” These were analyzed to ensure the presence of voicing during the closure phase of the [g] and an absence of voicing during [p] (for an illustrative spectrogram, see Figure 1).

The duration of medial vowels presented in licit tokens was measured. On average, /i/ duration in licit tokens was 66 ms (range 60–73 ms, $SD = 5$ ms) and /u/ duration was 67 ms (range 60–74 ms, $SD = 6$ ms). These are slightly longer than measurements provided by a study examining Japanese speaker vowel duration in /hVba/ nonce words (/i/ = 47 ms, /u/ = 51 ms; Nishi et al., 2008).

2.1.4 Predictions. Experiment 1 tests the following hypotheses.

H1) Japanese listeners will categorize the majority of illicit ([eCpo]) tokens as the equivalent licit string with medial /u/ (i.e., /eCupo/), but those illicit tokens where the first consonant has low Entropy and low Surprisal in /Ci/ sequences (i.e., /ɛʃpo/ and /etʃpo/) will more frequently be assigned to the medial /i/ category (/eCipo/) than those with high Surprisal (/egpo/). Furthermore, due to the difference in the predictability (Entropy) of low Surprisal CV contexts (/ʃi/ vs. /tʃi/), /etʃpo/ will be more frequently assigned to the medial /i/ licit category than /ɛʃpo/. If so, this will provide evidence that the collocation frequency of two phones in CV sequences modulates the vowel quality of epenthetic vowels. This pattern is predicted to also be observed in goodness-of-fit scores whereby the acceptability of illicit tokens will vary according to the relative transitional probability of the consonant and vowel sequence.

H2) Phonotactically licit sequences that are transitionally improbable will occasionally be assigned to categories that represent sequences that are highly probable. If true, this would show that phonotactically conditioned misperception is not entirely modulated by the phonotactic rules of the listener's L1. The most likely phonotactically attested candidate for such misperception is the [etʃupo] token, because /tʃu/ has the highest Surprisal of all tested diphones while /tʃi/ has the lowest Surprisal (Table 3). In addition, the /tʃ/ consonant has the lowest Entropy of all consonants tested (Table 4).

H3) Participants who are more likely to assign /ɛʃpo/ tokens to the /ɛʃpo/ category will also be more likely to assign /etʃpo/ tokens to the /etʃpo/ category. If this is the case, this would show that certain individuals are more sensitive to transitional probability information than others, but nevertheless sensitivity to transitional probability is consistent across contexts.

2.2 Results

2.2.1 Categorization. Participants assigned most illicit /eCpo/ tokens to their predicted /eCupo/ category ($M = 74\%$; $SD = 11\%$). Of the three illicit tokens, [etʃpo] was most frequently assigned to its respective /i/-epenthesized category, /etʃipo/ ($M = 46\%$; $SD = 22\%$), followed by [ɛʃpo] to /ɛʃipo/ ($M = 31\%$; $SD = 19\%$), while there were barely any instances of [egpo] being categorized as /egipo/ ($M = 0.3\%$; $SD = 0.014\%$). These findings are compatible with H1, whereby the allocation of tokens to medial /i/ categories adheres to predictions based on transitional probability. This pattern is evident across both the Categorization experiment and the following Discrimination experiment (Experiment 2).

A repeated measures analysis of variance (ANOVA) revealed a significant difference between the categorization rates of the three illicit tokens, $F(2, 58) = 68.629$, $p < 0.001$. A post hoc multiple comparison with Bonferroni correction revealed a significant difference between all three illicit tokens categorization rates; [egpo]/[ɛʃpo] ($p < 0.001$), [egpo]/[etʃpo] ($p < 0.001$), and [ɛʃpo]/[etʃpo] ($p = 0.006$). Asymmetry in categorization rates adheres to predictions based on Surprisal and Entropy.

To further investigate these findings, illicit token categorization response times were analyzed. Participants took longest to categorize [etʃpo] ($M = 1574$ ms, $SD = 165$ ms) tokens,

followed by [eʃpo] ($M = 1519$ ms, $SD = 153$ ms) and finally [egpo] ($M = 1412$ ms, $SD = 174$ ms). A repeated measures ANOVA indicated that mean response times differed significantly between the three illicit tokens, $F(2, 58) = 19.632$, $p < 0.001$. Post hoc analysis revealed a significant difference between the response times for [egpo] tokens and both [eʃpo] ($p < 0.001$) and [etʃpo] ($p < 0.001$). However, there was no significant difference in the response times of the categorization of [eʃpo] and [etʃpo] tokens ($p = 0.105$). These findings suggest that in contexts where /i/ is expected and/or /u/ is unexpected, perceptual systems require more time to process input and resolve ambiguities.³

The present study suggests that phonotactically conditioned perceptual repair is not modulated by phonotactic constraints alone, but also by transitional probability. This would suggest that even phonotactically attested sequences of speech might undergo assimilation to another string provided that the signal is suitably improbable and is perceptually similar enough to a highly probable target. The rate at which phonotactically attested tokens were assigned a category where the medial vowel does not match to the token (e.g., [etʃupo] to /etʃipo/) was therefore analyzed. Consider [etʃupo]: /tʃu/ has the highest Surprisal of all tested diphones (and /tʃi/ has the lowest Surprisal: Table 3) and /tʃ/ has the lowest Entropy of all consonants (Table 4). This token was miscategorized most often; every participant assigned [etʃupo] to the /etʃipo/ category (8%) on at least one occasion of the 27 times they were exposed to this phonotactically licensed token. A repeated measures ANOVA calculated on the number of times each participant assigned a phonotactically licensed token to an unexpected category returned a significant result, $F(5, 145) = 58.259$, $p < 0.001$ (Table 6). A subsequent post hoc multiple comparison with Bonferroni correction revealed a significant difference between the rate that [etʃupo] was assigned to /etʃipo/ and all other miscategorization results ($p < 0.001$ in all cases). No other contrasts were significant. This suggests that phonotactically conditioned illusions are not exclusively motivated by the listeners' intrinsic understanding of phonotactic constraints.

It is important to note the absence of a similar miscategorization effect on [eʃupo] tokens, which were very rarely categorized as /eʃipo/ (<1%), where /ʃu/ has a low Surprisal and /ʃ/ has the

Table 6. The results of post hoc multiple comparisons for miscategorization rates.

		Mean diff.	Std. error	p
[etʃupo] to /etʃipo/	[etʃipo] to /etʃupo/*	1.767	0.248	<0.001
	[egupo] to /egipo/*	2.033	0.162	<0.001
	[egipo] to /egupo/*	2.267	0.143	<0.001
	[eʃupo] to /etipo/*	1.433	0.282	<0.001
	[eʃipo] to /eʃupo/*	2.133	0.157	<0.001
[etʃipo] to /etʃupo/	[egupo] to /egipo/	0.267	0.203	1
	[egipo] to /egupo/	0.5	0.171	0.1
	[eʃupo] to /eʃipo/	-0.333	0.268	1
	[eʃipo] to /eʃupo/	0.367	0.162	0.476
	[egupo] to /egipo/	0.233	0.124	1
[egupo] to /egipo/	[eʃupo] to /eʃipo/	-0.6	0.207	0.105
	[eʃipo] to /eʃupo/	0.1	0.154	1
	[egipo] to /egupo/	-0.833	0.284	0.097
[egipo] to /egupo/	[eʃupo] to /eʃipo/	-0.133	0.079	1
	[eʃipo] to /eʃupo/	0.7	0.304	0.429

* indicate a significant difference between categorization rates.

second lowest Entropy of tested consonants. In fact, the rate that [ɛjipo] tokens were categorized as /ɛjupo/ (3%) was higher than the rate that [ɛjupo] tokens were categorized as /ɛjipo/. However, this relatively high rate is largely due to two participants who assigned [ɛjipo] tokens to /ɛjupo/ five and six times, respectively. The absence of miscategorization of [ɛjupo] tokens may be due to a threshold effect whereby phonotactically licit sequences require an especially low transitional probability in order to motivate assimilation.

To test H3—some individuals are more sensitive to transitional probability information than others—a simple linear regression analysis was calculated on each participant's average categorization rate of the [ɛjpo] token to the /ɛjipo/ category ($M = 31\%$; $SD = 19\%$) based on [ɛtjpo] to /ɛtjipo/ categorization rates ($M = 46\%$; $SD = 22\%$). A significant correlation was found, $F(1, 29) = 5.353$, $p = 0.046$, with an R^2 of 0.135 (Figure 2). This is an interesting finding because it reveals that individual variation in sensitivity to transitional probability is, at least to some extent, consistent across contexts.

2.2.2 Goodness of fit. While the goodness-of-fit scores assigned to illicit tokens are largely consistent with the categorization pattern and response times, they differ in some key areas. Average goodness-of-fit scores are presented in parentheses in Table 7. On average, [egpo] and [ɛjpo] tokens achieved higher average goodness-of-fit scores when they were assigned to the corresponding /u/ category ([egpo] to /ɛgupo/: 3.89; [ɛjpo] to /ɛjupo/: 4.93) than their respective /i/ category ([egpo] to /ɛgipo/: 3.33; [ɛjpo] to /ɛjipo/: 4.92). However, [ɛtjpo] tokens that were assigned to /ɛtjupo/ (4.5) achieved lower goodness-of-fit ratings than those assigned to /ɛtjipo/ (4.63). It is important to note that the [egpo] token was only assigned to the /ɛgipo/ category on three occasions and these data points have been omitted from the following analysis. In addition, one participant

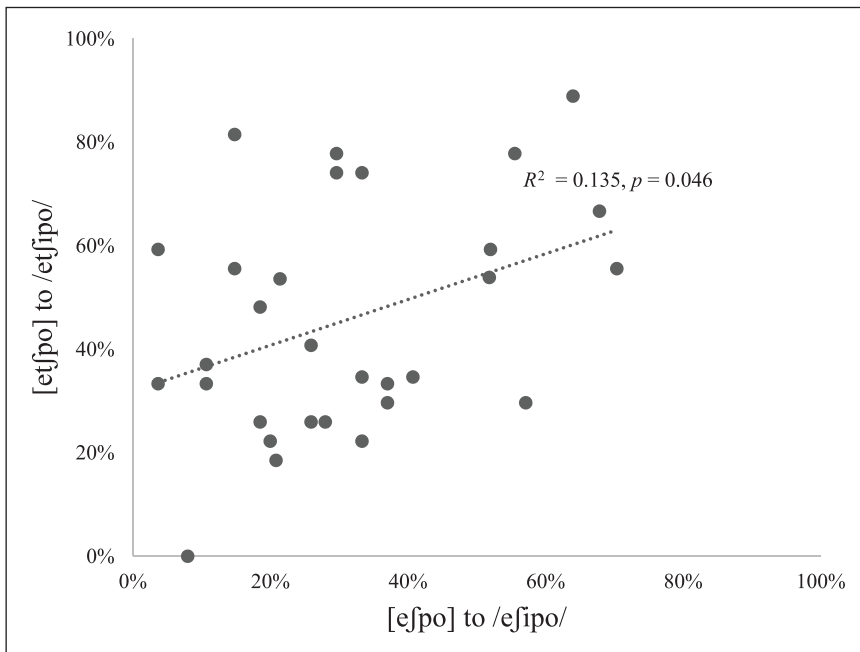


Figure 2. Scatterplot revealing the relationship between rates at which [ɛtjpo] and [ɛjpo] were assigned to their respective /ɛCipo/ categories by individual participants.

Table 7. Categorization rates and goodness-of-fit scores, averaged across all responses. Categorization patterns: Tokens appear vertically in [square] brackets, Categories appear horizontally in /slashes/. Figures not presented in parentheses represent the percentage of responses to a given category and figures in parentheses represent the average goodness-of-fit ratings on a scale from 1 (different) to 7 (identical). Goodness-of-fit scores for results less than 1% are not included.

	/egipo/%	/egupo/%	/ejipo/%	/ejupo/%	/etjipo/%	/etjupo/%
[egipo]	99 (5.48)	1 (2.29)				<1
[egpo]	<1 (3.33) ^a	100 (3.89)				
[egupo]		100 (5.63)		<1	<1	
[ejipo]			97 (5.34)	3 (3.84)		
[ejpo]			31 (4.92)	69 (4.93)	<1	
[ejupo]			<1	99 (5.42)		<1
[etjipo]	<1		<1		98 (4.98)	2 (2.80)
[etjpo]		<1	<1		46 (4.63)	54 (4.50)
[etjupo]				<1	8 (5.04)	92 (5.19)

^aThe [egpo] token was only assigned to the /egipo/ category on three occasions during Experiment 1.

recorded zero instances of assigning the [etjpo] token to the /etjipo/ category and therefore provided no goodness-of-fit score. A repeated measures ANOVA with average goodness-of-fit scores (excluding [egpo] to /egipo/) as the dependent variable and categories as the independent variable was calculated. The repeated measures ANOVA revealed a significant effect of illicit tokens on the goodness-of-fit score, $F(4, 116) = 14.205$, $p < 0.001$. However, a post hoc multiple comparison with Bonferroni correction revealed no significant difference between [ejpo] to /ejupo/ and [ejpo] to /ejipo/ goodness-of-fit scores ($p = 1$) or [etjpo] to /etjupo/ and [etjpo] to /etjipo/ goodness-of-fit scores ($p = 1$). This finding is not supported by H1, which proposes that both categorization rates and goodness-of-fit scores are modulated by transitional probability.

2.3 Experiment 1: Discussion

Experiment 1 reveals a clear difference in the likelihood that Japanese listeners will perceive an epenthetic /i/—rather than a /u/—across the three illicit consonant conditions. Categorization rate, categorization response time, and goodness-of-fit scores all suggest that cluster onset /tʃ/ is most likely to motivate /i/ perceptual epenthesis, followed by /ʃ/, while /g/ does not appear to elicit /i/ perceptual epenthesis (or does so extremely infrequently). To account for the observed patterns, we propose to use varying levels of Surprisal and Entropy, which are calculated on biphone transitional probabilities. These findings are predicted by Entropy calculations (Section 1.2), which were calculated on the transitional probabilities of high vowels. While both /tʃ/ and /ʃ/ occur more frequently with /i/ than with /u/, /tʃ/ has a much lower Entropy (0.25) than /ʃ/ (0.36) because the following vowel is more predictable in the case of /tʃ/. Importantly, the relatively high categorization rate of the [etjupo] token to the /etjipo/ category suggests that even phonotactically licit sequences can undergo perceptual assimilation provided that the input is suitably improbable and that the target is highly predictable. This is important because it implies that phonotactically conditioned repair is not necessarily motivated by phonotactic constraints but rather by the predictability of the co-occurrence of sounds in a listener's L1. However, the lack of [ejupo] to /ejipo/ categorization suggests that either the input must have a certain level of improbability or the target must have a certain level of predictability (most likely a combination of the two) in order to motivate perceptual repair. Here, we present this as evidence to support our proposal that speech

perception, in particular the quality of perceptually epenthesis vowels, is modulated by L1 transitional probability, especially perhaps at the limits of the system. Lastly, the within-participant correlation between assigning [eʃpo] to the /eʃipo/ and [etʃpo] to the /etʃipo/ category suggests that certain individuals are more sensitive to transitional probability information than others.

3

Experiment 2: Discrimination experiment

3.1 Method

3.1.1 Participants. The participants were the same 31 native speakers of Japanese who completed Experiment 1. As noted in Section 2.1.1, one participant was unable to successfully complete the experiment and their data was subsequently discarded from the analyses.

3.1.2 Procedure. Participants completed Experiment 2 following a 5 minute break after the completion of Experiment 1. Experiment 2 involved 288 AXB discrimination trials, which tested the discriminability of /egpo/–/egipo/, /egpo/–/egupo/, /eʃpo/–/eʃipo/, /eʃpo/–/eʃupo/, /etʃpo/–/etʃipo/, and /etʃpo/–/etʃupo/ contrasts. To avoid sequence bias, the individual tokens for each AXB triad were constructed so that the tokens were counterbalanced and occurred in every position of the triad: each triad comprised one token from each of the three speakers (1, 2, 3) organized into six sequences (123, 132, 213, 231, 312, 321) and each of these sequences was organized into four different triad sequences (AAB, ABB, BAA, BBA: where the letters denote a stimulus token). During the experiment, each AXB discrimination trial was drawn twice at random from this array of 24 triads (six speaker sequences times four triad types). Since we included six contrasts (shown in Table 8), this resulted in a total of 288 trials. To increase task demand so as to avoid a potential ceiling effect, all trials were presented to participants in a single block at random. The AXB discrimination task was presented on a laptop using *Psyscope X* (Bonatti, 2010) and discrimination responses and response times were recorded automatically by the application. Response times were recorded from the beginning of the third token of each triad. We collected response times because they are not subject to the same ceiling effects as AXB task accuracy and can provide an additional and more accurate indication of task difficulty. Upon being presented with the triad, participants would push either the 1 or the 3 button on the keyboard to indicate which they believed was a better match to the middle token. The AXB discrimination triads were spaced with a 1500 ms inter-trial interval. Within AXB trials, the inter-stimulus interval was 1000 ms to ensure phonological rather than phonetic processing (Werker & Tees, 1984). If participants failed to respond to trials within a 2000 ms response window, they were presented with an on-screen message in Japanese, politely asking them to respond more quickly. Missed trials were then replayed at a random time during the remainder of the experiment. As with Experiment 1, participants were asked to complete a practice task prior to beginning Experiment 2. The task consisted of perceptually distinct tokens in the A and B positions and an exact copy of either the A or B token (drawn at random)

Table 8. Contrasts for AXB tests.

Condition	/i/ Epenthesis	/u/ Epenthesis
/g/	/egpo/–/egipo/	/egpo/–/egupo/
/ʃ/	/eʃpo/–/eʃipo/	/eʃpo/–/eʃupo/
/tʃ/	/etʃpo/–/etʃipo/	/etʃpo/–/etʃupo/

in the X position. A researcher monitored the practice task to ensure that participants understood the instructions.

3.1.3 Stimuli. The stimuli used in Experiment 2 were the same /eCVpo/ and /eCpo/ strings used in Experiment 1 above. However, the discrimination experiment only uses the third of the five repetitions produced by the AusE speakers. Medial /i/ and /u/ in /eCVpo/ strings both had an average of 66 ms.

3.1.4 Predictions. In Experiment 2, we test the following hypotheses.

H4) Overall, participants will be less accurate at discriminating between /eCupo/–/eCpo/ contrasts than /eCipo/–/eCpo/ contrasts. As in Experiment 1 (Section 2), previous studies have shown that for Japanese listeners, /u/ is the default epenthetic vowel (e.g., Dupoux et al., 2011), so participants will have more difficulty distinguishing illicit tokens from licit tokens with a medial /u/. In addition, this accuracy difference will also be reflected in the response times of individuals, whereby contrasts with poorer discrimination accuracy will display longer average reaction times from participants.

H5) Consonants with a high transitional probability with /i/ (/tʃ/ and /ʃ/) will be more likely to elicit /i/ perceptual epenthesis than /g/—which has a low transitional probability with /i/. This will be evident in AXB test accuracy, where we predict that listeners are less accurate at discriminating between low Entropy /eCipo/–/eCpo/ contrasts. In addition, we also predict that listeners will take longer to respond to low Entropy /eCipo/–/eCpo/ contrasts due to the increased difficulty of these contrasts.

H6) Participants who were more likely to assign illicit tokens to /eCipo/ categories in Experiment 1 (Section 2) will achieve a lower accuracy when discriminating between /eCipo/ and /eCpo/ tokens. If correct, this would provide further evidence that certain individuals are more likely to rely on transitional probability than others.

3.2 Results

Overall, participants more accurately discriminated between /i/ contrasts (e.g., /egipo/–/egpo/; $M = 93\%$; $SD = 8\%$) than /u/ contrasts (e.g., /egupo/–/egpo/; $M = 82\%$; $SD = 14\%$), suggesting that /u/ is, indeed, the default candidate for perceptual epenthesis despite higher /i/ transitional probability with /tʃ/ and /ʃ/. However, the difference in discrimination accuracy between the /tʃ/ and /ʃ/ conditions reveals an asymmetry that reflects differences in Entropy. For example, the difference between contrasts /eʃipo/–/eʃpo/ ($M = 77\%$; $SD = 11\%$) and /eʃupo/–/eʃpo/ ($M = 72\%$; $SD = 12\%$) is 5%, while the difference between /etʃipo/–/etʃpo/ ($M = 73\%$, $SD = 13\%$) and /etʃupo/–/etʃpo/ ($M = 72\%$; $SD = 11\%$) contrasts is only 1%. The existence of multiple epenthetic vowels suggests that /u/ and /i/ perceptual epenthesis in the tested contexts are in competition, while the perceptual system selects between probable combinations of phones. Therefore, differences in accuracy and response times in /u/ and /i/ discrimination tests that share a consonantal environment were also tested (represented by Acc diff. and RT diff. in Table 9).

A repeated measures ANOVA calculated on each participant's average accuracy for each of the six discrimination tests showed a significant effect of contrast on accuracy, $F(5, 145) = 43.693$, $p < 0.001$. A post hoc multiple comparison with Bonferroni correction revealed a significant difference between the accuracy of the /egipo/–/egpo/ contrast and all other contrasts ($p < 0.001$) and the /etʃipo/–/etʃpo/ contrast and all other contrasts ($p < 0.001$) except for /eʃipo/–/eʃpo/

Table 9. AXB discrimination test accuracy and response time differences. Included within are the standard deviations for accuracy (Acc SD) and response time (RT SD) as well as the differences in accuracy in tests that share a preceding phonemic environment (Acc diff.) and the difference in response times (RT diff.).

AXB test	Accuracy (%)	Acc SD (%)	Acc diff. (%)	RT (ms)	RT SD (ms)	RT diff. (ms)
/egipo/–/egpo/	93	8	11	1040	102	–117
/egupo/–/egpo/	82	14		1157	141	
/eʃipo/–/eʃpo/	77	11	5	1160	107	–42
/eʃupo/–/eʃpo/	72	12		1202	107	
/etʃipo/–/etʃpo/	73	13	1	1201	102	5
/etʃupo/–/etʃpo/	72	11		1196	109	

($p = 0.074$). No other contrasts differed significantly. To test H5—which proposes that contrast accuracy is contingent on the transitional probability of the medial vowel in phonotactically attested tokens with its preceding consonant—the discrimination accuracy differences between contrast pairs that share a preceding consonant need to be examined. These values are presented in the Acc diff. columns in Table 9. A second repeated measures ANOVA was calculated on each participant's average accuracy difference between contrasts that share a preceding consonant (e.g., /egipo/–/egpo/ accuracy subtracted from their /egupo/–/egpo/ accuracy). The second repeated measures ANOVA showed a significant effect of the preceding consonant on accuracy difference, $F(2, 58) = 8.567$, $p < 0.001$. A post hoc multiple comparison with Bonferroni correction conducted on the second repeated measures ANOVA revealed a significant difference between /g/ contrasts and both /ʃ/ ($p = 0.034$) and /tʃ/ contrasts ($p = 0.002$), but not between /ʃ/ and /tʃ/ contrasts ($p = 0.37$).

As with Experiment 1, response times were analyzed, which were recorded by button press. A repeated measures ANOVA calculated on each participant's average response time for each of the six discrimination tests showed a significant effect of contrast on response time, $F(5, 145) = 25.602$, $p < 0.001$. A post hoc multiple comparison with Bonferroni correction revealed a significant difference between the response time of the /egipo/–/egpo/ contrast and all other contrasts ($p < 0.002$). Response time difference between other contrasts did not achieve significance. As with calculations conducted on the accuracy differences between contrasts that share a preceding consonant, a second repeated measures ANOVA calculated on each participant's average accuracy difference was conducted (see RT diff. column in Table 9). The second repeated measures ANOVA showed a significant effect of the preceding consonant on response time, $F(2, 58) = 15.921$, $p < 0.001$. A post hoc comparison on the second repeated measures ANOVA revealed a significant difference between /g/ contrasts and both /ʃ/ ($p = 0.001$) and /tʃ/ contrasts ($p < 0.001$), but not between /ʃ/ and /tʃ/ contrasts ($p = 0.112$). AXB discrimination accuracy and response times reveal a clear influence of the three consonant conditions on Japanese perceptual epenthesis. These are used as a proxy to measure perceptual epenthesis, under the assumption that lower accuracy rates and higher response times are indicative of stronger illusory epenthetic vowels.

To determine whether there was a relationship between participants' categorizations in Experiment 1 and their discrimination accuracy in Experiment 2 (H6), a linear regression was fit to examine the correlation between between /eCipo/ categorization rate of illicit /eCpo/ tokens and the accuracy of participants in /eCipo/–/eCpo/ trials (Figure 3). The correlation was significant, $F(1, 89) = 27.845$, $p < 0.001$, with an R^2 of 0.24. However, a second simple linear regression calculated on the reverse—the correlation between the categorisation rate of illicit /eCpo/ tokens into /eCupo/ and /eCpo/–/eCupo/ discrimination rate—did not achieve significance,

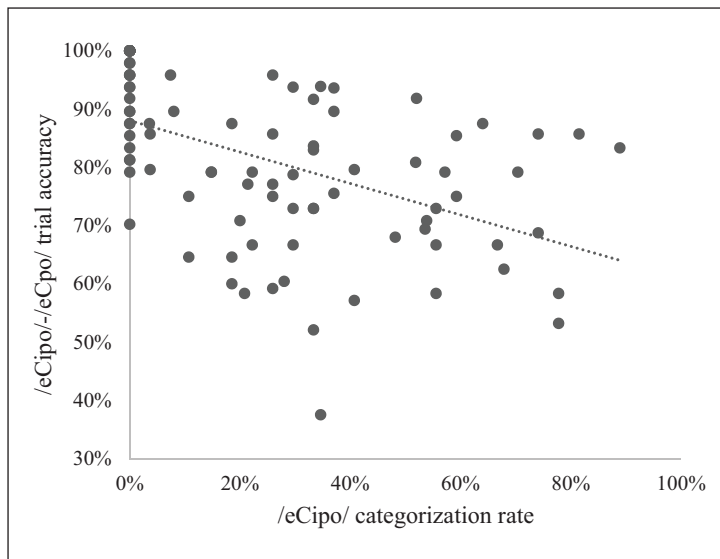


Figure 3. Scatterplot revealing the negative relationship between illicit tokens categorized to /eCipo/ categories and /eCipo/-/eCpo/ trial accuracy.

$F(1, 89) = 2.67, p = 0.106$, with an R^2 of 0.029. These results may suggest that, unlike /i/-perceptual epenthesis, which we claim is modulated by transitional probability, /u/ is simply a default candidate for perceptual epenthesis, occurring in all but a few suitably predictable conditions.

3.3 Discussion

The findings of Experiment 2 are consistent with those of Experiment 1 where the illusion of /i/ perceptual epenthesis appears to be stronger following /tʃ/, followed by /ʃ/ and finally /g/. This is evident in both discrimination accuracy and the response times of AXB tasks. Participants were least accurate at discriminating between /eCpo/ tokens and /eCipo/ tokens when the preceding consonant would predictably be followed by an /i/ in Japanese discourse (i.e., following /tʃ/). In Experiment 2, we propose that the differences between the /tʃ/ and /ʃ/ conditions are the result of the differential transitional probability in these two conditions. Conditions where both /Ci/ Surprisal and Entropy are low (e.g., following /tʃ/) achieved lower /eCipo/-/eCpo/ AXB trial accuracy than conditions where /Ci/ Surprisal is low but Entropy is less so (e.g., following /ʃ/). This suggests that for a listener to perceive /i/ rather than the default epenthetic /u/, it is important that the preceding consonant is most likely followed by a suitable alternate epenthetic candidate (as measured by Surprisal) and that other vowels occur at a low frequency, making the context predictable (as measured by Entropy).

Overall, participants were more accurate at discriminating between /eCipo/-/eCpo/ contrasts than /eCupo/-/eCpo/ contrasts, suggesting a propensity for /u/ perceptual epenthesis. This is despite /i/ occurring more frequently (thus having a lower Surprisal) than /u/ after /tʃ/ and /ʃ/ in CV sequences. Despite the overall higher discrimination of /eCipo/-/eCpo/ contrasts, we observe a very clear influence of Surprisal and Entropy on the quality of epenthetic vowels, just one that is not impactful enough to completely override the “dominant” vowel, /u/, to the point where Japanese listeners experience /i/ perceptual epenthesis the majority of the time. The trials that

proved to be the most difficult to distinguish were /etʃupo/–/etʃpo/ ($M = 72.29\%$; $SD = 11\%$) and /eʃupo/–/eʃpo/ ($M = 72.30\%$; $SD = 12\%$), which yielded almost identical accuracy results. The least difficult trials were /egupo/–/egpo/ ($M = 81.69\%$; $SD = 14\%$) and /egipo/–/egpo/ ($M = 93.04\%$; $SD = 8\%$), which 11 out of 30 participants were able to successfully distinguish 100% of the time. We propose that the reason for the high accuracy scores on the /egupo/–/egpo/ pair (compared to the other pairs), is due to the fact that the environment following voiced stops in Japanese is not an environment that conditions vowel devoicing for native speakers. Hence, there is no situation in Japanese in which a sequence [gp] would ordinarily correspond to a phonemic (conceptual) string /gVp/. This is unlike the case with the two voiceless sequences involving /ʃ/ and /tʃ/, where surface sequences such as [ʃp] and [tʃp] do, often, correspond to strings such as /ʃip/, /ʃup/, /tʃip/, and /tʃup/.

Finally, those participants who were more likely to assign illicit tokens to their corresponding /eCipo/ category in Experiment 1 were also more likely to achieve a lower accuracy score for /eCipo/–/eCpo/ discrimination trials (Figure 3). This provides further evidence that some participants hear /i/ perceptual epenthesis more frequently and more robustly than other participants across contexts. This is an important finding because it further strengthens our assertion that some listeners are more sensitive to transitional probability than others.

4 Conclusion

The results from Experiment 1 (categorization) and Experiment 2 (discrimination) reveal that the perception of epenthetic vowels in Japanese perceptual epenthesis is modulated by transitional probability. We deployed two measures of predictability (Surprisal and Entropy) of given sequences that have previously been shown to have a significant influence on the robustness with which speakers produce speech (e.g., Aylett & Turk, 2004, 2006; Bell et al., 2009; Cohen Priva, 2012, 2015; Hume, 2016; Shaw & Kawahara, 2017; Zipf, 1948). Our findings suggest that there is a relationship between the co-occurrence frequency of sounds in a listener's linguistic experience and the epenthetic target. Both experiments provide evidence that /tʃ/ is most likely to elicit /i/ perceptual epenthesis, followed by /ʃ/, and finally /g/. This pattern is exhibited in the categorization rates and response times in Experiment 1 and in the discrimination accuracy and response times in Experiment 2. Surprisal calculations alone predict that /ʃ/ should elicit /i/ perceptual epenthesis more than /tʃ/ (see Table 3). Entropy is therefore an important factor in determining the quality of epenthetic vowels. Surprisal must, nevertheless, be an important factor in these findings because participants experienced /i/ perceptual epenthesis in sequences where /Ci/ has low Surprisal, and not in those (as in the environment following /g/) where Surprisal of /i/ is high. One possible interpretation of these findings is that listeners rely upon Surprisal to varying extents depending on the level of Entropy. This means that when the vowel that follows the consonant is relatively more predictable (Entropy is low), listeners are more likely to rely on transitional probability information; on the other hand, when the vowel is relatively less predictable (Entropy is high), listeners are more likely to resort to the default epenthetic /u/ regardless of Surprisal.⁴

In Experiment 1, none of the three /eCpo/ tokens were assigned more often to the /eCipo/ category than the /eCupo/ category. In Experiment 2, /eCpo/–/eCipo/ triads achieved a higher average discrimination accuracy than /eCpo/–/eCupo/ triads. While these differences were not significant for the /etʃpo/ tokens, this suggests that /i/ was unable to surpass /u/ as the dominant epenthetic vowel despite /i/ occurring more frequently than /u/ in /ʃV/ and /tʃV/ contexts in our corpus. However, this does not mean that the dominant epenthetic vowel cannot be superseded. Research has shown that Korean listeners are significantly more likely to perceive an alternate epenthetic vowel (/i/) when the production of the dominant vowel (/u/) would violate Korean phonotactics

(Durvasula & Kahng, 2015). Such phonotactic restrictions would limit Korean listeners' exposure to these sequences to near zero and reduce the Surprisal and Entropy of the alternate epenthetic conditions. While the present study does not provide conclusive evidence that alternate epenthetic vowels can predictably be experienced in certain contexts, we propose that this is possible, provided the alternate epenthetic vowel is appropriately minimal and the context is suitably predictable and probable. A likely additional factor in determining the quality of epenthetic vowels is the Surprisal of the default candidate for perceptual epenthesis—that which is most perceptually minimal—which theoretically would need to have a high enough level of Surprisal for alternate epenthetic vowels to be considered for perceptual epenthesis.

The present study suggests that listeners tune to the transitional probabilities of their L1. The miscategorization rate of the phonotactically licit sequence [etʃupo] to the /etʃipo/ category suggests that even phonotactically licit sequences can undergo assimilation provided that the signal is suitably improbable. While further exploration of this finding is needed, it suggests that transitional probability modulates speech perception at the segmental level. Studies have shown that humans “tune” to the sounds of their L1 throughout infancy (e.g., Best, 1994; Best et al., 1995; Kuhl et al., 2006; Werker & Tees, 1984); our findings suggest that transitional probability should be considered as one of the factors involved in such tuning.

Both experiments presented here provide evidence that some speakers rely more on transitional probability information than others. In Experiment 1, for instance, those participants who were more likely to categorize /eʃpo/ as /eʃipo/ were also more likely to categorize /etʃpo/ as /etʃipo/. Similarly, those participants who were more likely to assign /eCpo/ tokens to /eCipo/ categories in Experiment 1 were less accurate at discriminating between /eCpo/ and /eCipo/ tokens in Experiment 2. This raises an interesting question as to why certain individuals are more attuned to transitional probability information than others. While the specific features of the Japanese /u/ that make it the default candidate for perceptual epenthesis are still not entirely clear, it is likely that it involves allophones of /u/ having a lack of acoustic cues resulting from shortened duration, lower intensity, devoicing/deletion, or centrality. It is also possible that the transitional probability of /u/—either measured against all vowels, high vowels, or devoiced vowels—contributes to its default status. Regardless, the finding that certain individuals are more likely to experience the alternate epenthetic vowel should not be framed as a top-down/bottom-up distinction, where one type of processing relies on linguistic experience while the other relies on acoustic information, because both types rely on linguistic experience.

An alternate interpretation of these results is that transitional probability is not responsible for the variance across the three contexts examined, and that perceptual epenthesis is simply modulated by the frequency at which vowels undergo devoicing. While vowel devoicing rates have been shown to influence the frequency and/or saliency of /u/ perceptual epenthesis (Kilpatrick et al., 2018), this does not adequately explain our findings. Both Japanese high vowels undergo devoicing more frequently when the preceding consonant is a fricative than when the preceding consonant is an affricate (Maekawa & Kikuchi, 2005). The near identical discrimination accuracy of /etʃupo/–/etʃpo/ ($M = 72\%$; $SD = 11\%$) and /eʃupo/–/eʃpo/ ($M = 72\%$; $SD = 12\%$) trials suggests that our findings are not the result of devoicing rates following these two consonants.

A second possible alternate interpretation is that these results can be explained by differences in Surprisal alone, without any reference to Entropy. If Entropy is calculated against all vowels—rather than high vowels only—then Entropy for /tʃ/ and /ʃ/ becomes much more similar (0.85 and 0.83, respectively), while /g/ retains the highest Entropy (1.58). Under such an interpretation, Surprisal is a better candidate for explaining our findings because /tʃu/ (8.75) has a higher Surprisal than /ʃu/ (4.91). In the present study, we hypothesize that Entropy—alongside Surprisal—modulates phonotactically conditioned repair because production studies have shown that Entropy, along with

Surprisal, is a good predictor of speaker behavior (see the discussion in Shaw & Kawahara, 2017). In addition, Entropy was calculated against high vowels, because Entropy measures based on high vowels have been found to be an accurate predictor of Japanese vowel devoicing (Whang, 2018) and studies have shown a clear relationship between vowel devoicing and Japanese perceptual epenthesis (Kilpatrick et al., 2018). Despite this, we remain open to the possibility that Surprisal may play a more substantial role than in the current proposal. Nevertheless, we note that even under this alternative interpretation, our general conclusion—some Information Theoretic predictability measures, whether they be Surprisal or Entropy or both—remains the same.

The present paper employs calculations that provide a measure of informativity based on co-occurrence frequency (Surprisal) and a measurement of the uncertainty of the vowel that follows each of the consonants (Entropy). These two calculations may not account for all the findings of our experiments. For instance, it is possible that the overall frequency at which each vowel occurs in Japanese discourse influences the quality of epenthetic vowels. However, in the CSJ-RDB (Maekawa, 2003), /u/ occurs least frequently (12%) of all short vowels, while /i/ is the third least frequent (20%). Another issue is that Entropy cannot be calculated on sequences beyond the diphone due to a lack of /etʃu/ and /tʃup/ sequences. Even if the corpus contained a small number of these sequences, the usefulness of calculating Entropy on such small numbers is questionable, because these sequences are rarely present in the speech experience of Japanese listeners.

In conclusion, the present paper demonstrates that the quality of Japanese epenthetic vowels can be influenced by the preceding consonant. We propose that this is due to the variance in transitional probability of the CV sequences at issue here. The findings of Experiments 1 and 2 support our central hypothesis that listeners are tuned to some predictability measures in their L1. When the incoming signal is suitably improbable, it is sometimes misperceived as that which is probable. The significant difference between perceptual epenthesis in /etʃpo/ and /eʃpo/ tokens suggests that Entropy acts as a regulator for transitional probability information whereby listeners take transitional probability information into account to a greater extent when sequences are relatively more predictable (have lower Entropy). This suggests that phonotactically conditioned linguistic illusions such as perceptual epenthesis are the result of erroneous predictions made based on the listener's linguistic experience. To make these predictions, perceptual systems consider perceptual similarity, co-occurrence frequency (Surprisal), and predictability (Entropy) of L1 sounds to try to match unfamiliar sequences to somewhat similar, familiar, and predictable sequences. Given that no /eCpo/ context exhibited a preference for /i/ perceptual epenthesis, we propose that perceptual similarity is the most important factor in these predictions but argue that transitional probability nevertheless has a clear role in the perception of epenthetic vowels.

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Notes

1. To refer to the posterior coronal affricates, the present study uses postalveolar symbols ([tʃ] and [dʒ]), while Mattingley et al. (2015) use alveopalatal symbols ([tɕ] and [dʑ]). They refer to the same set of sounds.
2. This is true of “Standard Japanese”; certain dialects—particularly those found in the Kansai region—may behave differently.
3. We note that categorization responses were recorded by mouse click. All six categories were presented to participants regardless of the token being categorized (as opposed to just the relevant /eCupo/ and /eCipo/ categories) and categories were always arrayed in the same way on the screen. The standard practice for measuring response time in experiments such as those presented in the present paper is button press (as is the case in Experiment 2). Mouse click response times are typically not reported due to several issues, such as cursor starting point variability. However, given that the repeated measures ANOVA detected a significant effect ($p < 0.001$) and that the pattern observed is consistent with that observed in categorization rates and goodness-of-fit scores in Experiment 1 and discrimination accuracy and response times in Experiment 2, they are included in the present analysis.
4. An anonymous reviewer pointed out that there was only one condition with high Entropy (/g/), which was tested against lower Entropy /j/ and /tʃ/. Future studies should address the generalizability of the current findings.

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